

PROJECT TEMPLATE

AI WEATHER WISE

Weather Information Application

Project Name	AI Weather Wise
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Six-page topic-based project template

Edit the technology, API provider, and final features to match your actual project.

01. PROJECT INTRODUCTION

Project Title

AI Weather Wise

Introduction

AI Weather Wise is a weather information project designed to help users check weather conditions for a selected city or location. It presents weather details in a simple and understandable format.

Purpose

The purpose of this project is to make weather information easy to access so users can plan their daily activities, travel, and outdoor events.

Problem Statement

People may need to check weather details from different sources. This project aims to provide key weather information in one place through a weather data API.

Proposed Solution

The user enters a city name. The application requests weather data from an API and displays the available details, such as temperature, humidity, wind speed, and weather condition.

02. OBJECTIVES AND FEATURES

Main Objectives

- Show weather details for a selected location.
- Provide clear and easy-to-read information.
- Reduce the effort needed to check current weather.
- Display suitable messages when data is unavailable.
- Create a simple, user-friendly interface.

Key Features

- City/location search
- Current weather display
- Temperature and weather description
- Humidity and wind details, if supported by the API
- Loading, validation, and error messages

Target Users

Students, travellers, families, and anyone who wants to check weather information before planning daily activities.

Project Scope

The initial version focuses on searching a location and displaying available weather data. Additional features such as multi-day forecasts or saved cities can be added later.

03. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or laptop for development
- Internet connection to access the weather API
- Smartphone or computer browser for testing, if the application is web-based

Software Requirements

- Operating system: Windows, Linux, or macOS
- Code editor such as Visual Studio Code
- Web browser
- HTML, CSS, and JavaScript or another team-selected technology
- Weather API service

Functional Requirements

The system should accept a location, request weather data, process the response, and display the available weather details. It should also handle empty or invalid location input.

Non-Functional Requirements

The interface should be easy to use, responsive, reasonably fast, and reliable when the network and API service are available.

Note

Update this section if your team uses a different programming language, framework, API provider, or hardware setup.

04. SYSTEM DESIGN AND WORKFLOW

System Workflow

1. User opens the AI Weather Wise application.
2. User enters a city or location.
3. The application checks the input.
4. A request is sent to the selected weather API.
5. The API returns available weather data.
6. The application displays the result.

Simple Architecture

User → Search Interface → Application Logic → Weather API → Weather Result

Main Modules

- Location Search Module
- Input Validation Module
- Weather API Request Module
- Data Processing Module
- Weather Display Module
- Error Handling Module

Interface Plan

The main screen may include the project title, a city search box, a search button, and a result area for weather information. The final design should match the team's implementation.

05. IMPLEMENTATION AND TESTING

Implementation Steps

1. Select and review a weather API.
2. Prepare the interface layout.
3. Connect the application to the API.
4. Read the response data.
5. Display the relevant weather fields.
6. Add validation and error handling.
7. Test and improve the application.

Test Cases

- Valid city: weather details should be displayed.
- Empty input: ask the user to enter a location.
- Unknown city: show a clear error message.
- No internet: show a connection message.
- API service error: handle the error without crashing.

Expected Output

The application is expected to show available weather details for the chosen location in a clear format. Exact fields depend on the API service selected.

Safety and API Key

Follow the API provider's usage rules. Avoid placing private API keys in public code, and do not collect unnecessary personal information.

06. EXPECTED OUTCOME AND FUTURE SCOPE

Expected Outcome

AI Weather Wise is expected to provide a convenient way to view weather details for a selected location. It can also help the team learn API integration, interface design, data handling, and software testing.

Limitations

The result depends on internet access, API availability, supported locations, update frequency, and any usage limits imposed by the API provider.

Future Enhancements

- Multi-day weather forecast
- Weather icons and improved visual design
- Saved favourite cities
- Responsive mobile interface
- Optional weather alerts, if supported

Conclusion

AI Weather Wise is a useful project concept for presenting weather information through an API. With a simple interface and proper error handling, it can make weather details easier for users to understand.

References

Add the official documentation of the weather API and any frameworks or tools actually used by the team.